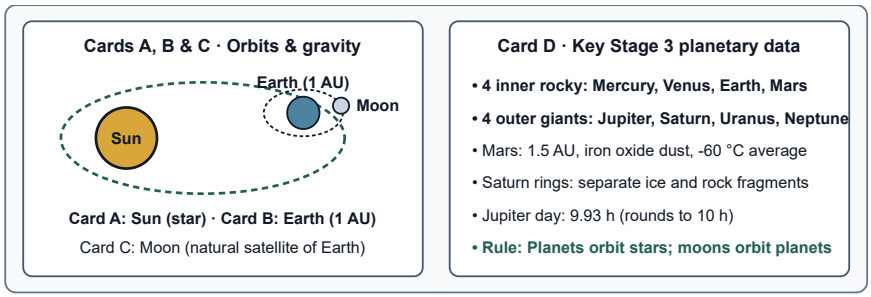
Reference sheets

Reference A · Class Solar System Source Cards (Cards A, B, C, and D)

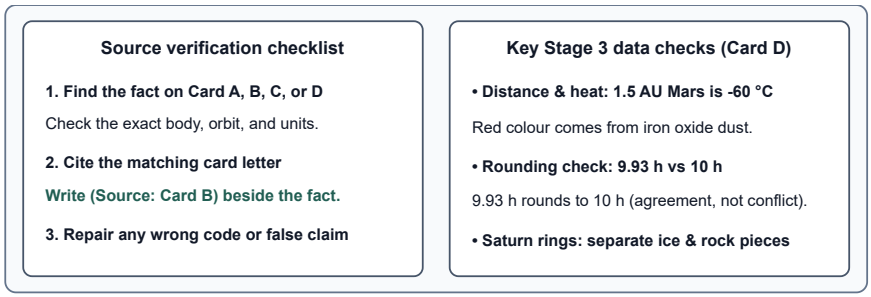


Class Source Cards A, B, C, and D showing the Sun's gravity, Earth's 1 AU orbit, the Moon's satellite orbit, and Key Stage 3 planetary data.

Reference only. Do not write on this sheet. Read the source card first, then cite its exact code beside your fact.

Source card code	Body and classification	Verified science facts on this card	Key Stage 3 mechanism or measurement
Card A · Star card	The Sun (central star)	The Sun is a hot star at the centre of our Solar System that emits its own light	The Sun's non-contact gravitational pull holds planets in curved orbits around it
Card B · Planet card	Earth (inner rocky planet)	Earth is a nearly spherical rocky planet that orbits the Sun once a year	Earth orbits at an average distance of 1 astronomical unit (1 AU, about 150 million km)
Card C · Moon card	The Moon (natural satellite)	The Moon is a rocky natural satellite that orbits Earth in about a month and reflects sunlight	Earth's gravity holds the Moon in orbit around Earth rather than directly around the Sun
Card D · Solar System data	4 inner rocky planets & 4 outer giants	Inner rocky: Mercury, Venus, Earth, Mars. Outer giants: Jupiter, Saturn, Uranus, Neptune	Mars (1.5 AU, -60 °C, iron oxide dust); Saturn rings (ice and rock fragments); Jupiter day (9.93 h, rounds to 10 h)

Reference B · Key Stage 3 source verification and claim repair guide



Key Stage 3 source verification guide: 1. Find the fact on Cards A to D, 2. Cite the card code, 3. Repair wrong citations or false claims.

Research check	Faulty or incomplete note	How to verify and repair using Cards A, B, C, and D
Missing source code	'Earth orbits the Sun at 1 AU.' (no source named)	Add the matching card code so any reader can check it: 'Earth orbits the Sun at 1 AU (Source: Card B).'
Wrong source code	'The Moon orbits Earth (Source: Card A).'	Card A is about the Sun. Replace the code with Card C: 'The Moon orbits Earth (Source: Card C).'
False planet claim	'The Moon is a planet because it is round.'	Cards B, C, and D show that planets orbit a star (the Sun), whereas the Moon orbits a planet (Earth) as a natural satellite.
False temperature or ring claim	'Mars is red so it is hot; Saturn has a solid ring.'	Card D shows Mars is red from iron oxide dust and cold (-60 °C at 1.5 AU), and Saturn's rings are separate ice and rock pieces.
Rounded data comparison	Table 1 gives Jupiter's day as 9.93 h; Table 2 gives 10 h.	Card D confirms both agree after rounding: 9.93 hours rounds to 10 hours to the nearest whole hour.

Made by Matt

Tier pack · Independent

Use Reference A and Reference B. Work in your book or on this sheet. Point, speak, sign, draw, type, write, or tell an adult. You may pass.

Arrival · 4 questions

1. Use the arrival picture cue. Which body emits its own light, and which source card panel shows this?

Help: Check the yellow light rays on the left panel of the arrival drawing.

2. Compare the middle panel (Card B) and the right panel (Card C). How does a planet's orbit differ from a moon's orbit?

Help: Compare what sits at the centre of the orbit on Card B versus Card C.

3. Why is a science claim only an unchecked guess if it has no source card named beside it?

Help: Think about how scientists verify research notes.

4. A pupil writes: 'The Sun is a star (Source: Card C).' Explain what is faulty in this note.

Help: Compare the heading on Card A with the heading on Card C.

You do tasks

1. Use Reference A. Write two verified Solar System facts about different bodies and cite the exact class source card (Card A, Card B, Card C, or Card D) for each fact.

Staff I ☐ S ☐ C ☐

Help: Include both the scientific fact and (Source: Card ___) for each sentence.

2. A poster claims: 'The Moon is a planet because it is spherical and moves through space.' Use Card B, Card C, and Card D on Reference A to repair this claim. **Staff I ☐ S ☐ C ☐**

Help: Compare what a planet orbits (Card B and Card D) with what the Moon orbits (Card C).

3. Use Card D on Reference A to check and repair two faulty notes: (a) 'Mars is red because it is hotter than Earth', and (b) 'Saturn's rings are one solid metal band.'

Staff I ☐ S ☐ C ☐

Help: Check the Mars row (1.5 AU, iron oxide, -60 °C) and the Saturn row (ice and rock fragments) on Reference A and Reference B.

Optional participation-loop prompt: Share one verified fact and source code with an adult, hear one feedback step, and respond below or with // in your book. (Staff: the pupil responds to feedback with an adult, and R is marked only after the pupil has responded.)

Exit ticket

1. Write one verified Solar System fact and cite the exact class source card that backs it.

2. Why can Card A support a claim about the Sun, but not a claim about the Moon's orbit?

3. Repair this false claim and cite your source cards: 'The Moon is a planet.'

Pupil voice

What helped you today, and what would you change?

Point, speak, sign, draw, write or choose. You may pass.

Feedback and response · Independent

Feedback codes

Staff I ☐ S ☐ C ☐ · Route card feedback boxes: VF ☐ · WS ☐ · I ☐ · E ☐ · NS + … ☐ · R ☐ · // ☐ · ? ☐

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

My response: Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt